Defining and quantifying motion

WORKED SOLUTIONS — CLASS QUESTIONS AND PRACTICE

Only open these solutions after completing your classwork. Check and correct your work. Homework answers are not included.

Multiple choice (1–8)

1	2	3	4	5	6	7	8
B	C	C	A	C	C	C	B

Free response (9–14)

9. Average speed — Journey 1

1. **Equation:** Average speed = total distance / total elapsed time
2. **Rearrange:** Already arranged for average speed.
3. **Substitute:** Average speed = 30 m / 6 s
4. **Answer:** Average speed = 5 m/s. Speed alone does not specify direction, so velocity cannot be determined.

10. Displacement and average velocity — Journey 2

Displacement

1. **Equation:** $\Delta x = x_f - x_i$
2. **Rearrange:** Already arranged for Δx.
3. **Substitute:** $\Delta x = -8\ \text{m} - (+12\ \text{m})$
4. **Answer:** $\Delta x = -20$ m.

Average velocity

1. **Equation:** $v_{avg} = \Delta x / \Delta t$
2. **Rearrange:** Already arranged for v_{avg}.
3. **Substitute:** $v_{avg} = (-20\ \text{m}) / (5\ \text{s})$
4. **Answer:** $v_{avg} = -4$ m/s. The minus sign means motion in the negative direction.

11. Uniform acceleration — Journeys 3 and 4

Journey 3: acceleration

1. **Equation:** $a = (v_f - v_i) / \Delta t$
2. **Rearrange:** Already arranged for a.
3. **Substitute:** $a = (+8\ \text{m/s} - 0\ \text{m/s}) / (4\ \text{s})$
4. **Answer:** $a = +2\ \text{m/s}^2$.

Journey 3: average velocity

1. **Equation:** $v_{avg} = (v_i + v_f) / 2$ (constant a)
2. **Rearrange:** Already arranged for v_{avg}.
3. **Substitute:** $v_{avg} = (0\ \text{m/s} + 8\ \text{m/s}) / 2$
4. **Answer:** $v_{avg} = +4$ m/s.

Journey 3: displacement

1. **Equation:** $\Delta x = v_{avg}\ \Delta t$
2. **Rearrange:** Already arranged for Δx.

3. **Substitute:** $\Delta x = (+4\ \text{m/s})(4\ \text{s})$
4. **Answer:** $\Delta x = +16\ \text{m}$.

Journey 4: acceleration

1. **Equation:** $a = (v_f - v_i) / \Delta t$
2. **Rearrange:** Already arranged for a.
3. **Substitute:** $a = (0\ \text{m/s} - 8\ \text{m/s}) / (4\ \text{s})$
4. **Answer:** $a = -2\ \text{m/s}^2$.

Journey 4: displacement

1. **Equation:** $\Delta x = [(v_i + v_f)/2]\ \Delta t$ (constant a)
2. **Rearrange:** Already arranged for Δx.
3. **Substitute:** $\Delta x = [(8\ \text{m/s} + 0\ \text{m/s})/2]\ (4\ \text{s})$
4. **Answer:** $\Delta x = +16\ \text{m}$. Both cars travel the same displacement in the same time, but their accelerations are opposite.

12. Slowing down — Journey 5

Acceleration

1. **Equation:** $a = (v_f - v_i)/\Delta t$
2. **Rearrange:** Already arranged for a.
3. **Substitute:** $a = (6\ \text{m/s} - 10\ \text{m/s})/(4\ \text{s})$
4. **Answer:** $a = -1\ \text{m/s}^2$.

Displacement

1. **Equation:** $\Delta x = [(v_i + v_f)/2]\ \Delta t$ (constant a)
2. **Rearrange:** Already arranged for Δx.
3. **Substitute:** $\Delta x = [(10\ \text{m/s} + 6\ \text{m/s})/2] \times 4\ \text{s}$
4. **Answer:** $\Delta x = +32\ \text{m}$. The car slows because acceleration is opposite to its positive velocity.

13. Reversing direction — Journey 6

Time when the car first stops

1. **Equation:** $v_f = v_i + a\Delta t$
2. **Rearrange:** $\Delta t = (v_f - v_i)/a$
3. **Substitute:** $\Delta t = (0\ \text{m/s} - 4\ \text{m/s})/(-2\ \text{m/s}^2)$
4. **Answer:** $\Delta t = 2\ \text{s}$.

Final velocity after 4 s

1. **Equation:** $v_f = v_i + a\Delta t$
2. **Rearrange:** Already arranged for v_f.
3. **Substitute:** $v_f = 4\ \text{m/s} + (-2\ \text{m/s}^2)(4\ \text{s})$
4. **Answer:** $v_f = -4\ \text{m/s}$.

Net displacement after 4 s

1. **Equation:** $\Delta x = v_i\ \Delta t + \frac{1}{2}a(\Delta t)^2$
2. **Rearrange:** Already arranged for Δx.
3. **Substitute:** $\Delta x = (4\ \text{m/s})(4\ \text{s}) + \frac{1}{2}(-2\ \text{m/s}^2)(4\ \text{s})^2$
4. **Answer:** $\Delta x = 0\ \text{m}$.

Total distance

1. **Equation:** Distance = distance outward + distance back
2. **Rearrange:** Use the stopping instant to split the journey.
3. **Substitute:** Distance = 4 m + 4 m
4. **Answer:** Total distance = 8 m (although displacement is zero).

14. Outward and return journey

Total distance

1. **Equation:** Distance = path length 1 + path length 2
2. **Rearrange:** Already arranged for distance.
3. **Substitute:** 40 m + 10 m
4. **Answer:** Distance = 50 m.

Displacement

1. **Equation:** Displacement = signed change in position
2. **Rearrange:** Take east as positive.
3. **Substitute:** $\Delta x = +40\ \text{m} - 10\ \text{m}$
4. **Answer:** $\Delta x = +30$ m (east).

Average speed

1. **Equation:** Average speed = total distance / total time
2. **Rearrange:** Already arranged for average speed.
3. **Substitute:** 50 m / 10 s
4. **Answer:** Average speed = 5 m/s.

Average velocity

1. **Equation:** $v_{avg} = \Delta x / \Delta t$
2. **Rearrange:** Already arranged for v_{avg}.
3. **Substitute:** (+30 m)/(10 s)
4. **Answer:** $v_{avg} = +3$ m/s (east).